

Intermediate Macroeconomic Theory

Econ 4002, Washington University in St. Louis

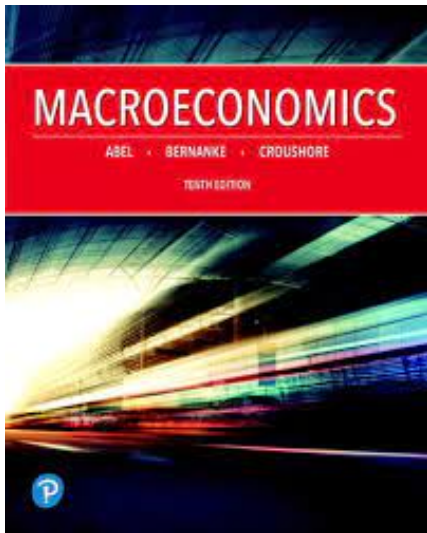
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Federal Reserve Bank of St. Louis

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This Course: Essential Info

- ▶ Econ 4002/5002, Intermediate Macroeconomic Theory
- ▶ Lectures: Mondays and Wednesdays, 4-5:20pm, Louderman Hall 00458
- ▶ 28 lectures, including this one, two review classes, and one in-class midterm exam
- ▶ One final exam
- ▶ **Assistants in Instruction:** Rain Cao and Jiayao Huang
- ▶ **Textbook:** *Macroeconomics, 11th Edition* (Abel, Bernanke & Croushore), Pearson

Textbook: very low resolution pictures of the cover



This Course: Essential Info

- ▶ **Grading:** 9 homeworks (approximately weekly) + 1 midterm exam + 1 final exam

$$\text{Final grade} = 0.3 \times \text{Homeworks} + 0.35 \times \text{Midterm} + 0.35 \times \text{Final}$$

- ▶ Homeworks cover roughly one set of lectures
- ▶ Typically assigned on Wednesdays, due the following Wednesday at 5:20 PM
- ▶ First homework drops on September 2
- ▶ Collaboration allowed, but individual submissions required
- ▶ Lowest HW grade dropped for final grade
- ▶ **Exams**
 - ▶ Scheduled for October 21 and December 11 (closed-book, calculators only)
 - ▶ Location: midterm exam is in class (4-5:20pm), final is location TBD (6-8pm)
 - ▶ Let me know ASAP if you cannot make it

This Course: Essential Info

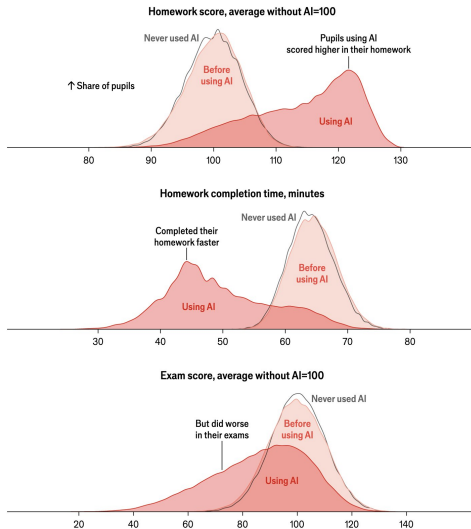
▶ **TAs and Office Hours**

- ▶ Rain Cao (c.langrui@wustl.edu)
Office hours: 2-3pm, Tuesdays, Seigle 372
- ▶ Jiayao Huang (hjiayao@wustl.edu)
Office hours: 2-3pm, Mondays, Seigle 377

▶ **Prerequisites**

- ▶ Econ 1502 and Econ 4001
- ▶ I assume you are familiar with basic statistics and differential calculus (i.e., optimization)
- ▶ Level of math in the lectures/homeworks goes beyond that of the book
- ▶ All of this info (and more) in the syllabus available on Canvas
- ▶ Lecture slides will be under “Discussions”

AI can help you do the homework - but does it help you learn?



Source: "The generative AI learning penalty: evidence from Chinese secondary education", by D. Strömberg et al., preprint June 2026

*26,811 pupils aged 12-18

Course Plan

1. Measurement (Chs. 1, 2)
2. Theory (Chs. 4, 5, 6, 7) - Midterm exam
 - ▶ Productivity, Output, and Employment
 - ▶ Consumption, Saving, and Investment
 - ▶ Growth
 - ▶ Assets, Money, and Prices
3. Analysis (Chs. 8, 9, 10, 11, 12, 14, 15) - Final exam
 - ▶ Business Cycles
 - ▶ A general macro model
 - ▶ Classical analysis
 - ▶ Keynesian analysis
 - ▶ Unemployment and inflation
 - ▶ Monetary policy
 - ▶ Fiscal policy (if we have time)

Learning Objectives

By the end of this course you should (I hope):

1. Have a good grasp of what the main macroeconomic aggregates are and how they are measured
 - ▶ i.e., GDP, inflation, government spending, interest rates, labor force participation
2. Understand the meaning of changes in those aggregates
 - ▶ i.e., “real GDP grew by an annualized 3% this quarter”, “year-on-year inflation was 5.4% last month”, “labor force participation fell to an historic low of 61% during the COVID-19 pandemic”
3. Critically evaluate macroeconomic issues using real-world data
4. Apply macroeconomic models to interpret and understand how the economy responds to shocks and policies
 - ▶ What are the effects of government stimulus checks on the inflation rate?
 - ▶ Does it make sense for the Fed to raise rates right now?
 - ▶ What does this imply for output growth?

Learning Objectives

By the end of this course you should be able to understand what is going on in this figure, from a Deutsche Bank market research report that explains why inflation picked up after the COVID-19 pandemic

Figure 1: Present equilibrium is slightly deflationary - the supply curve has shifted and become vertical but the demand curve has shifted more

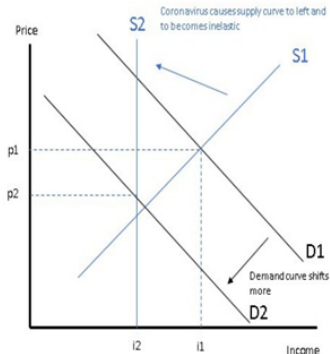
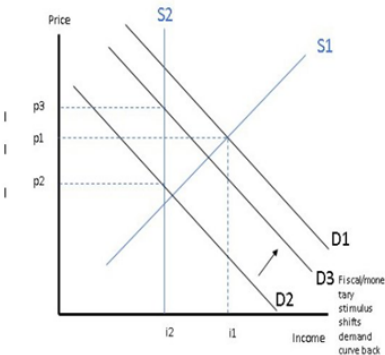


Figure 2: But fiscal and monetary stimulus will shift demand curve back towards pre coronavirus position, pushing up prices



Resources

- ▶ Wall Street Journal, Financial Times, Bloomberg News
- ▶ FRED: <https://fred.stlouisfed.org/>
- ▶ OECD: <https://data.oecd.org/>
- ▶ Eurostat: <https://ec.europa.eu/eurostat>
- ▶ Penn World Table: <https://www.rug.nl/ggdc/productivity/pwt/?lang=en>

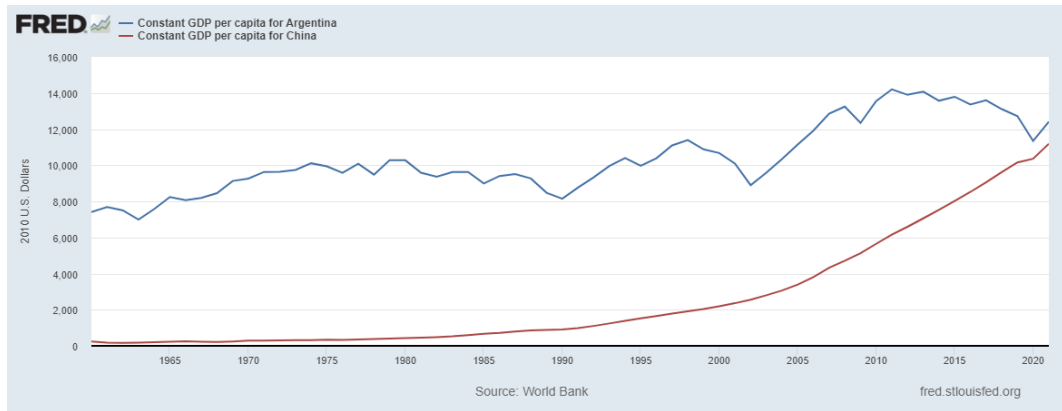
What is Macroeconomics?

$$\text{Economics} \simeq \text{Microeconomics} + \text{Macroeconomics}$$

- ▶ Why $\simeq$ instead of $=$?
 - ▶ Economics includes some fields and areas of study that are hard to classify
 - ▶ Macroeconomics increasingly overlaps with microeconomics
- ▶ Textbook definition: “the study of the structure and performance of national economies and of the policies that governments use to try to affect economic performance”
- ▶ Classical definition: macroeconomics encompasses the study of growth (long-run) and fluctuations (short and medium-run) of aggregate economies
- ▶ Easier to define it based on the questions it tries to answer:

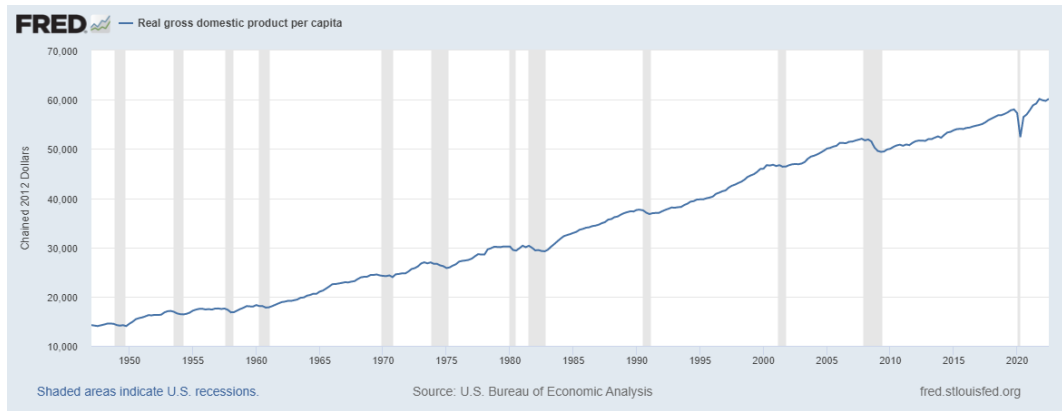
Why do some countries grow so fast?

...while others stagnate?



Why do economies experience business cycles?

...and what determines their length and intensity?



Real gross domestic product per capita

$$y_t = \frac{Y_t}{Pop_t}$$
$$Y_t = \frac{P_t Y_t}{P_{2012}}$$

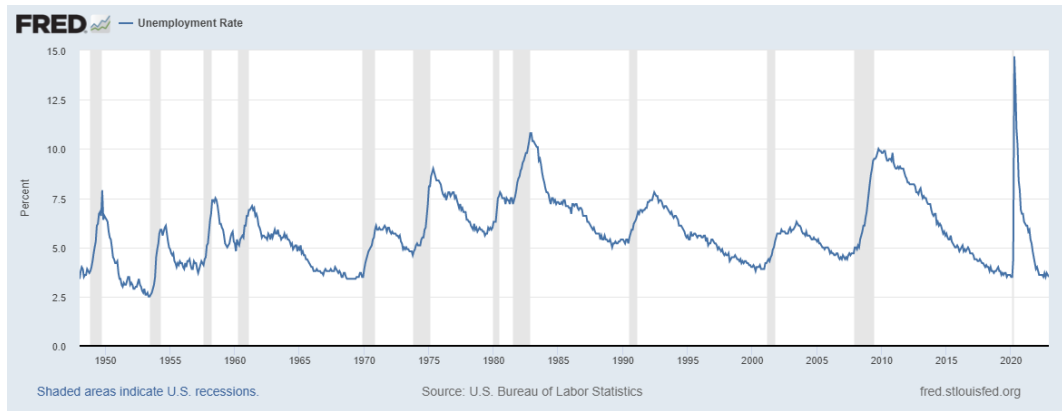
where

- ▶ Y_t = real GDP in year t
- ▶ P_t = price level in year t
- ▶ $P_t \times Y_t$ = nominal GDP in year t
- ▶ P_{2012} = price level in 2012
- ▶ Pop_t = population in year t

Measure of domestic production per person that is comparable over time, as it adjusts for changes in the price level.

Why are there unemployed people?

...even during expansions?



Unemployment Rate

$$u_t = \frac{U_t}{LF_t} = \frac{U_t}{U_t + E_t}$$

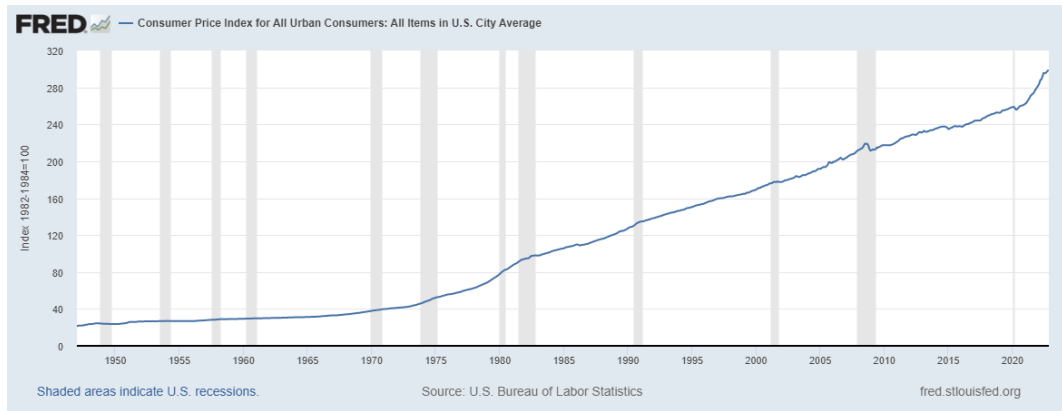
where

- ▶ U_t = number of unemployed people, people who are looking for a job and don't have one
- ▶ E_t = number of employed people
- ▶ $U_t + E_t = LF_t$ **labor force**, total number of people who are working or actively looking for a job
- ▶ Not everyone is in the labor force: children, retirees, military, full-time students, people not looking for a job, etc., $LF_t \neq Pop_t$
- ▶ Labor force participation rate:

$$LFPR_t = \frac{LF_t}{Pop_t}$$

Why do prices rise?

...and do they ever go down?



Consumer Price Index

$$P_t = \sum_{i=1} \omega_i P_{it}$$

where

- ▶ P_{it} = price of good i in period t
- ▶ ω_i = consumption weight of good i
- ▶ ω_i does not depend on t : allows us to compare the general price level of a fixed set of goods over time (in practice it is a bit more complicated than that!)

Inflation (month on month)

$$\pi_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

Inflation (year on year)

$$\pi_t^{\text{yoy}} = \frac{P_t - P_{t-12}}{P_{t-12}}$$

Can the government do anything about inflation and unemployment?

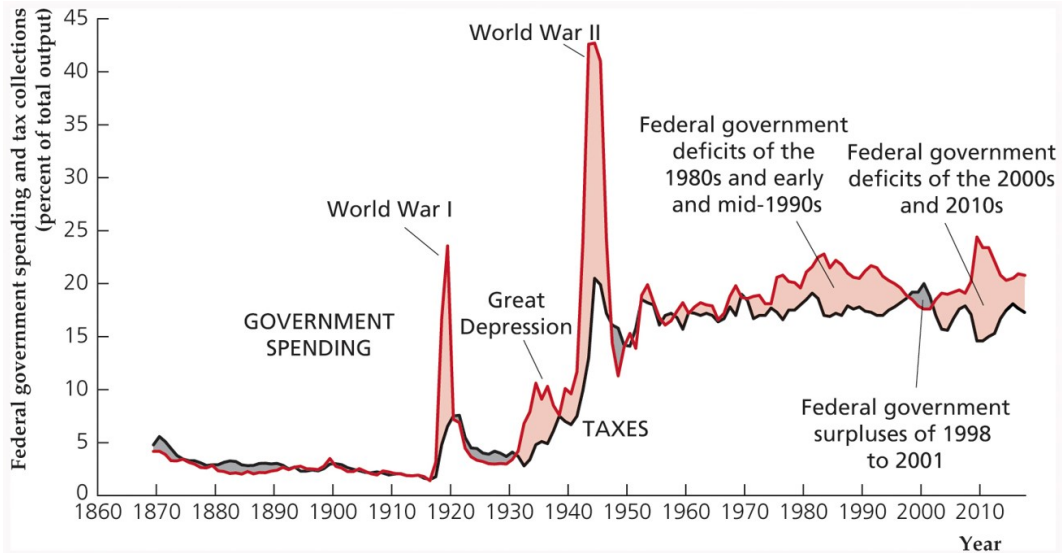
Fiscal Policy

- ▶ Relates to government spending, taxation, and the national debt
- ▶ Determined by the federal government, as well as state and local governments

Monetary Policy

- ▶ Relates to the money supply and interest rates
- ▶ Determined by the central bank
- ▶ The central bank of the United States is the Federal Reserve System (Fed)

US Government Spending and Revenue



Aggregation

- ▶ Aggregation underpins the key difference between microeconomics and macroeconomics
- ▶ Microeconomists traditionally study the individual decisions of economic agents, and in the equilibrium of individual markets
- ▶ Macroeconomics studies the aggregate effects of all such decisions, and how different markets are interconnected

Aggregation

- ▶ We will often talk about “aggregate output” or “aggregate consumption”

$$C_t = \sum_{i=1}^N c_{it}$$

- ▶ Aggregate consumption is the sum of individual consumption by all agents i in the economy, where N is the number of agents

$$c_{it} = \sum_{j=1}^J c_{jit}$$

- ▶ Individual consumption of agent i is the sum of consumption in all products and services j , where J is the number of consumption categories

What do macroeconomists do?

- ▶ Analysis
 - ▶ Private sector: financial institutions, consulting companies, technology companies, other large corporations, etc.
 - ▶ Public sector: Congressional Budget Office, White House Council of Economic Advisors, Federal Reserve, World Bank, International Monetary Fund, etc.
- ▶ Data development: Bureau of Economic Analysis, Bureau of Labor Statistics, Bureau of the Census, etc.
- ▶ Forecasting both in the public and private sectors
- ▶ Research
 - ▶ Universities
 - ▶ Policy institutions (i.e., central banks)
 - ▶ Think tanks

Positive vs. Normative Economics

Key distinction in economics, source of much disagreement among different schools of thought

- ▶ **Positive economics** (or descriptive) tries to describe reality as it is, developing and testing statements about the economy that are objective and verifiable
 - ▶ What are the effects of a tax cut on output?
 - ▶ What determines the demand for healthcare services?
 - ▶ What are the effects of a rise in interest rates on inflation?
 - ▶ Does a change in tariffs affect imports?
- ▶ **Normative economics** is based on prescriptive statements, developing statements that cannot be tested as they depend on someone's opinion
 - ▶ Should the government cut taxes?
 - ▶ Should the government provide free healthcare to all citizens?
 - ▶ Should the Fed raise interest rates?
 - ▶ Should the US impose tariffs on China?

Broad Learning Objective

I'd like you to leave this course knowing how to **think like an economist**:

1. Correlation does not imply causation. Examples:
 - ▶ Countries with more complex tax systems have higher income per capita
 - ▶ Government spending rises in recessions
2. Economic agents respond to incentives (people, firms, etc.)
 - ▶ Under appropriate conditions, competitive markets typically lead to efficient outcomes
 - ▶ Well-intended but poorly-designed policies can have very negative effects
3. Equilibrium effects matter
 - ▶ A change in behavior by a single agent (household, firm, etc.) can have very different consequences when everyone changes behavior at once